Nanosyntax

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nanosyntax

Abstract

Nanosyntax is a theory of syntax, morphology, and, in part, compositional semantics, which captures language variation as a consequence of different syntactic structures stored in the lexicon. The framework adheres to strict modularity, keeping the syntax free of both phonology and concepts. This is possible thanks to the absence of a pre-syntactic lexicon combined with (phrasal) lexicalisation after each syntactic operation. This entry discusses the basic tools of the framework, how it captures syncretisms and allomorphy, and how it accounts for language variation, but also how it seeks to keep to a sparse version of compositionality via a Gricean view of unary features.

Keywords: Nanosyntax, modularity, phrasal lexicalisation, unary features, containment, syncretism, allomorphy, suppletion, postsyntactic lexicon

Key Points

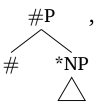
- In Nanosyntax, every terminal is a single (unary) feature, and hence every morpheme has an internal syntactic structure.
- Cross-linguistic variation is entirely driven by lexicalisation attempts.
- Nanosyntax is strictly modular: syntax is free of concepts and phonolology.

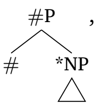
1 Introduction

This chapter lays out key notions of the framework of Nanosyntax. It starts out by presenting the basics of the framework, including the idea of a universal functional sequence, and phrasal lexicalisation (Sect. 2). Sect. 3 then shows how these ideas can be applied to account for syncretism and language variation resulting from cross-linguistic differences in the make-up of lexical entries. Sect. 4 extends the analysis to cases of allomorphy and suppletion in French verbal paradigms.

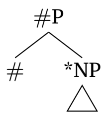
2 The basics

Nanosyntax (Starke 2002, 2009) is a theory unifying syntax, morphology, and to some extent compositional semantics. A core distinctive characteristic is that single morphemes are lexicalisations of entire syntactic trees. This results from the observation that every syntactic feature seems to have its distinct structural life and hence is its own syntactic terminal. That also entails that there is no pre-syntactic lexicon (understood as a language-specific list): lexical insertion is purely post-syntactic. This gives rise to a principled lexicon: the syntactic entries of lexical items (henceforth LIs) are always well-formed syntactic objects, i.e. a tree structure (as in (1) and (2)). LIs connect such stored structures with phonological information (in slanted brackets) and conceptual information (in small capitals). The LIs for *mouse* in (1) and *duck* (2) thus contain information linking different modules of the grammar (Caha 2022).

(1) < /maʊs/,  , MOUSE >

(2) < /dʌk/,  , DUCK >

These LIs store not only the features relevant to the noun phrase (with * indicating that a noun consists of more features than just N), but also a (singular) number feature (#). Languages other than English may lexicalise the singular feature with a separate morpheme, like Kipsigis *pata-yaan* 'duck-SG' (Kouneli, 2021; Blix, 2022).
The derivation of each clause, noun phrase, etc. starts from the relevant subset of a universal FUNCTIONAL SEQUENCE (henceforth FSEQ), needed to express the particular semantics of that clause, noun phrase, etc. Syntactic structures are built bottom-up by MERGE (Chomsky, 1995), grouping the features in the FSEQ into a binary branching structure. After every Merge, the syntax interfaces with the lexicon, searching for a candidate to lexicalize the phrasal node.

(3) 

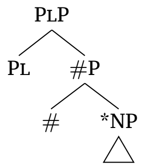
Both the LIs for *mouse* (1) and *duck* (2) can lexicalise (3), in accordance with the Identity Condition on Matching in (4) (Starke, 2002, 2009; Caha, 2009; De Clercq et al., To appear), also called the Superset Principle or Superset Effect (Starke 2009).

(4) A lexically stored constituent L matches a syntactic phrase S iff S is identical to L.

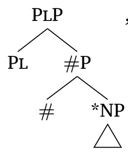
The **semantics** of a structure like (3) arises thanks to a **Gricean** logic (Grice 1978). #P in (3) represents a countable denotation containing both singularities and pluralities. However, if the plural feature [PL] is left out of the syntax, as in (3), the

meaning of the structure will be that of a singular. Put differently, if the strongest relevant information (PL) is not available, it can be inferred that this meaning is not relevant.
If PL is added to the derivation, (5), the semantics of the plural kicks in. In the case of the LI *mouse*, the LI (6) perfectly matches (5) and hence lexicalises it, overriding the previous lexicaliser, a process referred to as CYCLIC OVERRIDE.

(5)



(6) < /maɪs/,



, MOUSE >

During the lexicalisation of the singular structure (3), *mice* was also a candidate for lexicalisation, given the identity Condition on Matching (4): (3) is identical to a subconstituent both of the lexical tree for *mice* and that for *mouse*; compare (3) with both (1) and (6). There is thus a competition between the LIs *mouse* and *mice* for the lexicalisation of (3), resolved by the ELSEWHERE PRINCIPLE (Kiparsky, 1973), as formulated in (7) (Caha 2009, 55).

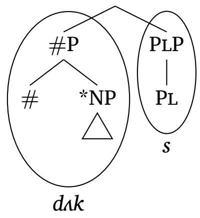
(7) In case two rules, R1 and R2, can apply in an environment E, R1 takes precedence over R2 if it applies in a proper subset of environments compared to R2.

As a result, *mouse* wins as the realisation of #P in the singular: its lexical tree applies in less environments than the tree for the plural form *mice*. An informal way of putting this is that *mouse* is a perfect match and has no unused features, compared to the lexical entry for *mice* (6), which has an unused feature PL when used for the singular form.
Consider what happens if the singular structure (3) is lexicalised by *duck*. The conceptual content of lexicalisations must remain fixed in a derivation, and hence 'mice' in (6) cannot override the singular in this case. The lexicon of English however contains another lexical item, i.e. the plural morpheme *-s*:

(8) < /s/, PLP >
PL

This entry does not match the base-generated structure of the plural in (5): no subtree of (5) is identical to the lexical tree in (8). This creates a problem since every operation must be followed by a (successful) lexicalisation.
In a bid to rescue the derivation, syntax performs evacuation movements, also called **lexicalisation-driven movement**, to help lexicalise PLP. Evacuation of #P leads to the structure in (9). In this case, evacuation helps: PLP can be lexicalized by (8) since the right branch of (9) is identical to the lexical tree (8).

(9)

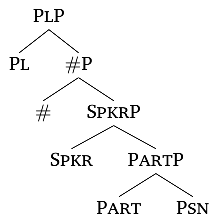


The order of operations attempted while building a tree is codified in a **Lexicalisation Algorithm** (Starke 2009, 2018, see also Caha et al. (in press); De Clercq et al. (To appear) for a detailed discussion of the algorithm and empirical case studies), or more recently simply referred to as the **Syntax Algorithm** as Nanosyntax increasingly derives traditional syntactic movements and constructions from its algorithm (see e.g. De Clercq (2020); Starke (2024)).

3 Syncretisms and language variation

A crucial part of the work done within Nanosyntax is the study of syncretisms, i.e. forms that can lexicalise several meanings or functions. A lot of research has been done in this area, like studies on the nominal domain (Caha 2009; Taraldsen 2010; Starke 2017; Bergsma 2019; Kloudová 2020; Türk & Caha 2021; Janků 2022), directional expressions (Pantcheva 2011), negation (De Clercq 2013, 2020; Baunaz & Lander 2023), demonstratives (Lander 2016), pronominals (Vanden Wyngaerd, 2018), complementizers (Baunaz 2018), cross-categorial syncretisms (Wiland, 2019), the inflectional and derivational morphology of verbs (Taraldsen Medová & Wiland 2019; Baunaz & Lander 2019; Starke & Cortiula 2021; Cortiula 2023), degrees (Caha et al. 2019; Vyshnevska 2022; Wiland 2023), and indefinites (Dekier 2021). In many empirical domains, syncretisms display the absence of ABA patterns, which points to a **containment** for the syntactico-semantic structures underlying these forms. In what follows, we consider some data from personal pronouns, also discussed in Vanden Wyngaerd (2018); Caha (2022). The containment hierarchy for personal pronouns is shown in (10), with the two number features we adopted earlier on top of the person hierarchy proper (as in Vanden Wyngaerd 2018). The first person is a person (PSN), which is not only a participant (PART) in the discourse, but also a speaker (SPKR), i.e. it consists of three features.

(10)



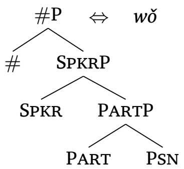
The second person is a person and a participant, but not a speaker; its structure is identical to (10) but without the feature SPKR. The third person is least complex in terms of its semantics and is just a person (Béjar 2003).
Support for the number features in the structure comes from the personal pronouns in Mandarin as in (11) (Corbett 2000, 76), which show that the singular forms are **contained** in the plural ones, suggesting that plural should be added to the structure of the singular, as in (10).

(11)

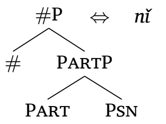
	SG	PL
1	wǒ	wǒ-men
2	nǐ	nǐ-men
3	tā	tā-men

Let us now show how the system handles language variation by assuming a universal FSEQ and variable lexical entries. Mandarin will have three different lexical items for singular person pronouns (12)-(14), but only one plural morpheme, as in (15), which will be added to the singular form as a consequence of lexicalisation-driven movement, as illustrated for the third person in (16).

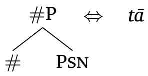
(12)



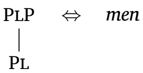
(13)



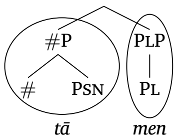
(14)



(15)



(16)

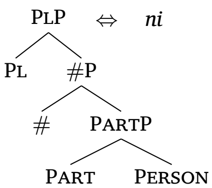


The system can also account for syncretism, as found in the pronomoninal system of Salt-Yui (17), where each person has a different form, which is syncretic between the singular and the plural. Each lexical item will thus contain a lexical tree that can lexicalise both the singular and the plural, as illustrated in (18) for 2nd person *ni*.

(17)

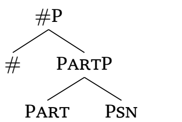
	SG	PL
1	na	na
2	ni	ni
3	DEM	DEM

(18)

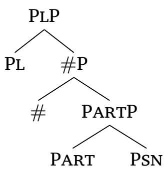


Thanks to the Matching Condition (4), this item can lexicalize the syntax of a 2nd person singular, (19) as well as plural (20).

(19)



(20)



The discussion of Mandarin and Salt-Yui personal pronouns shows that Nanosyntax provides a very concrete handle on **language variation**. Namely, it emerges as a consequence of which lexical trees are stored in the lexicon (Starke 2014). For the personal pronouns discussed in this entry, we focussed on how the size of an entry determines the variation. However, in some cases also the shape of the lexical tree matters. Lexical trees could for instance contain the result of movement. This type of stored lexical trees have been used to account for inverse marking patterns (Blix 2022), certain types of ABA-patterns in the verbal paradigm (Cortiula 2023), and complex patterns of stem allomorphy (Caha et al. 2024).

4 Allomorphy and suppletion

Nanosyntax uses phrasal lexicalisation not only to account for syncretisms, but also for allomorphy and suppletion. We illustrate the approach using suppletive verbal roots in French (Starke 2020, Starke 2022). Considering the present indicative (PRS), past indicative (PST) and the subjunctive (SBJV) in the singular, three different (suppletive) roots are used for a verb like *savoir* 'to know', as shown in (21). A regular verb like *fonder* 'to found' on the other hand makes use of just one root (22).[1]

(21)

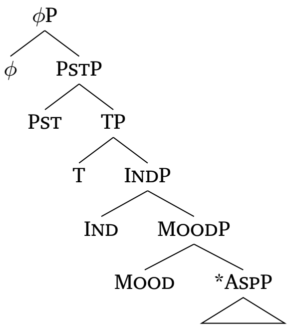
savoir	PRS	PST		SBJV	
1SG	sɛ	sav	ɛ	saʃ	ə
2SG	sɛ	sav	ɛ	saʃ	ə
3SG	sɛ	sav	ɛ	saʃ	ə

(22)

fonder	PRS		PST		SBJV	
1SG	fõd	ə	fõd	ɛ	fõd	ə
2SG	fõd	ə	fõd	ɛ	fõd	ə
3SG	fõd	ə	fõd	ɛ	fõd	ə

Starke (2020) proposes the FSEQ in (23).

(23)

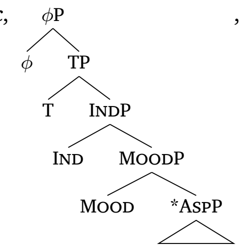


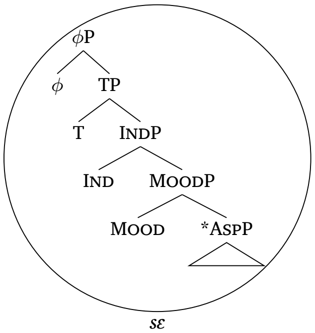
The highest features are the person- and number-related features taking care of agreement, represented here by means of the shorthand ϕ. The feature PST specifies the default tense feature (T) as past; its absence results in an interpretation as present, following the Gricean reasoning set out in Sect. 3. Similarly, IND on top of MOOD specifies the mood as indicative. The interpretation of MOOD in the absence of IND is subjunctive. The label *ASPP is shorthand for the features relevant to grammatical and lexical aspect, and whatever further features that make up the verb root (see Ramchand 2008).

We will now discuss the present, past, and subjunctive in turn, each time contrasting the irregular verb *savoir* with the regular *fonder*. French irregular verbs do not take an agreement ending

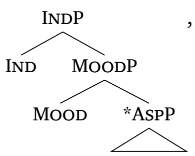
[1] In (21) and (22) the forms of the verbs are given in IPA, since the orthography of the present tense of *savoir* makes distinctions (*sais, sais, sait*) that do not exist in the spoken language.)

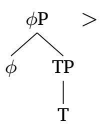
in the present singular. A minimal pair illustrating this is the irregular verb *fondre* 'to melt', which has the same root *fõd* and is hence sometimes homophonous with the regular verb *fonder* 'to found'. In the present singular, the lack of a suffix makes the irregular *fondre* come out as *fõ*, the final consonant remaining unprounced in the absence of a following vowel (Schane, 1966). In contrast, *fonder* comes out as *fõd* thanks to the presence of the agreement ending *ə*. We can thus conclude that irregular forms such as *sɛ* or *fõ* are capable of lexicalising the full verbal sequence on their own, requiring no agreement suffix in the present.

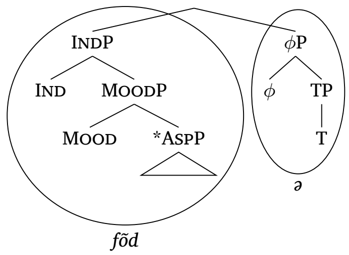
(24) < *sɛ*,  , KNOW >

(25) 

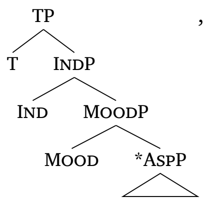
In contrast, the regular root *fõd* does not lexicalise all the features, which is why the agreement suffix *ə* appears. It is *a priori* unclear how many features in a form like *fõd-ə* are lexicalised by the root, and how many by the ending. However, the fact that *ə* is found both in the indicative and the subjunctive but not in the past suggests that it does not lexicalise either MOOD or IND, and that it does lexicalise the present tense feature T (and the feature above T, ϕ). This in turn entails that the root lexicalises all the features up to IND. The lexical entries are in (26a)-(26b), while the output of the syntactic derivation (which includes evacuation movement of INDP) is given in (26c).

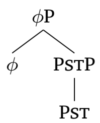
(26) a. < *fõd*,  , FOUND >

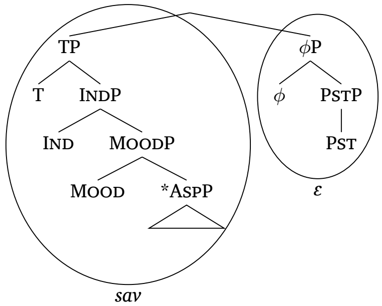
b. < *ə*,  >

c. 

What (24) and (26) illustrate is that Nanosyntax does not need to introduce special mechanisms to account for selection, i.e. why a particular root (the one of *fond-er* 'to found', but not the one of *fond-re* 'to melt', or the form *sɛ*) combines with a particular affix (*ə*): selection follows naturally from the structural size of morphemes in the lexicon. See Taraldsen (2010); Márkus (2015); Caha et al. (2019) for more details on the idea that root size determines affix distribution.
Consider next the past, which features a suppletive root in the case of *savoir* 'to know': *sav*; both regular and irregular verbs have the same suffix *ɛ*. The suppletion can be explained if the lexicon of French contains several different lexical items for *savoir*, each with slightly different features, but with the same conceptual information. Since the ending *ɛ* only occurs in the past, we can infer that it lexicalizes PST and ϕ, as in (27b). It follows that the root *sav* must be able to lexicalize all features up to T, including IND. Its entry is as in (27a). Root and suffix combine in the manner indicated in (28).

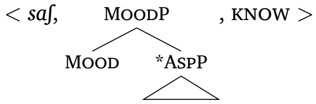
(27) a. < *sav*,  , KNOW >

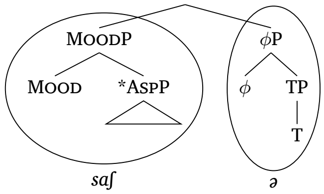
b. < *ɛ*,  >

(28) 

The regular root *fõd* in *fonder* 'to found' has the same size, and its derivation is identical to the one shown in (28) (modulo the different lexicalisation of TP).
The subjunctive of irregular *savoir* features yet another root (*saʃ*). Both regular and irregular verbs are followed by the same agreement ending *ə*. Given that *ə* starts at T (see (26b)), the root *saʃ* could in theory maximally reach up to IND. However, since this root is restricted to the subjunctive, we take its lexical tree to terminate at MOOD, as shown in (29). The root *saʃ* and the ending *ə* conspire to lexicalise the features of the subjunctive, as

indicated in (30). The regular verb *fonder* 'to found' works in the same way.

(29) 

(30) 

The above results are represented synoptically in Table 1. In this lexicalisation table, the column labels represent the FSEQ, the size of the various morphemes is indicated by the colouring (matching the one in (21) and (22) above), and absent features are marked by black cells. It shows the division of labour between the different roots and endings. The French lexicon stores three different roots for *savoir*, each with a different size, explaining why they appear in different syntactic derivations. Notice that while each of the suppletive roots is different, they represent a structural containment relation of the corresponding parts of the FSEQ. Only one root is stored for *fonder* 'to found', which can, however, lexicalise constituents of different sizes because of the Identity Condition on Matching in (4) above. This allows the theory to integrate both regular and irregular verbs into a single account.

Table 1 Lexicalisation table for PRS IND, PST, and SBJV of *savoir* 'to know' and *fonder* 'to found' (1-3P SG)

*ASPP	MOOD	IND	T	PST	ϕ
sɛ					
sav			ɛ		
saʃ			ə		
fõd			ə		
fõd			ɛ		
fõd			ə		

5 Conclusion

Nanosyntax provides an elegant way to capture language variation, as well as syncretism, allomorphy, and suppletion as a mere result of differences in the size of stored lexical items. It does so by making a number of minimal and partly independently needed assumptions, like the existence of MERGE, the FSEQ, evacuation movement, and phrasal lexicalisation.

Acknowledgments

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